

A Dual-Graph Counterfactual MAPPO Framework for Dynamic Multi-Robot Navigation

First A. Author, *Fellow, IEEE*, Second B. Author, and Third C. Author, Jr., *Member, IEEE*

Abstract—Multi-agent reinforcement learning (MARL) offers a principled way to coordinate autonomous ground vehicles (AGVs) in shared warehouse and logistics environments, yet two persistent obstacles limit its effectiveness in dense traffic. Shared team rewards provide only weak individual credit when agent interactions are tightly coupled, and deep integration of graph operators into the policy network introduces non-stationarity that destabilizes optimization. We propose a dual-graph counterfactual variant of multi-agent proximal policy optimization (MAPPO) that addresses both. A rolling-subgoal planner with adaptive detour logic decouples long-horizon routing from local conflict resolution. Two ego-centric graphs—one for social risk among neighboring agents, one for dynamic obstacle motion—each attend to a small set of high-risk entities ranked by predicted time-to-collision and motion descriptors, with a risk-biased graph attention network. A residual gate blends the resulting graph features into a stable multilayer-perceptron and long short-term memory (MLP-LSTM) navigation backbone only when interaction risk is elevated, preserving training stability in open spaces. A leave-one-out counterfactual critic estimates each agent's marginal contribution to team value and injects it into per-agent rewards before advantage estimation, sharpening cooperative credit without dual value heads or explicit value factorization. A partial-done training protocol with density-adaptive reward scaling preserves clean trajectory semantics across team sizes from two to eight robots. In Gazebo Circle-Swap evaluations, the framework improves success rate and minimum inter-agent separation over a backbone-only baseline, a fully connected graph baseline, and the classical optimal reciprocal collision avoidance (ORCA) baseline, with ablations confirming that each component contributes measurably to the final performance.

Index Terms—Multi-robot navigation, multi-agent reinforcement learning, graph attention networks, counterfactual credit assignment, curriculum learning.

I. INTRODUCTION

IN modern warehouse automation, logistics centers, and factory floors, autonomous ground vehicles (AGVs) operate in shared spaces to transport materials and sustain production throughput. Each vehicle must navigate from its current position to a dynamically assigned goal while avoiding static infrastructure, moving obstacles such as forklifts and human workers, and other vehicles pursuing their own goals. The difficulty is not merely geometric obstacle avoidance; it lies

in resolving strongly coupled interactions such as head-on encounters in narrow aisles, merging conflicts at intersections, and congestion in high-traffic zones under partial observability. As fleet density grows to meet throughput demands, small improvements in collision avoidance and coordination translate directly into operational uptime and safety margins.

Classical reactive methods such as reciprocal velocity obstacles (RVO) and optimal reciprocal collision avoidance (ORCA) [1], [2] are computationally efficient and interpretable, but they optimize only instantaneous local feasibility and cannot exploit interaction history. Deep reinforcement learning (DRL) jointly optimizes perception, local control, and multi-step interaction reasoning, with decentralized learners [3]–[5], light detection and ranging (LiDAR) based perception [6], [7], memory-augmented architectures [8], and path-constrained planning [9], [10] all advancing the state of the art [11]. Graph neural networks (GNNs) and attention mechanisms have been introduced to capture pairwise dependencies among agents [12]–[18]. Two challenges nevertheless persist. First, deep graph integration couples the evolving policy and the graph feature space, injecting non-stationarity that disrupts the trust-region updates of proximal policy optimization (PPO). Second, shared team rewards provide weak individual credit signals, so the centralized critic of multi-agent PPO (MAPPO) struggles to discriminate which agent's action caused a success or failure in tightly coupled events.

We address both issues with a dual-graph counterfactual MAPPO framework that separates stable local navigation from selective relational reasoning. Two lightweight ego-centric graphs—one for social risk and one for dynamic obstacles—each attend only to a small set of high-risk entities identified by predicted time-to-collision (TTC) descriptors. A residual fusion gate modulates a multilayer-perceptron and long short-term memory (MLP-LSTM) backbone only when conflict risk is elevated. A leave-one-out counterfactual critic estimates each agent's marginal contribution to team value and injects it into per-agent rewards before generalized advantage estimation (GAE), requiring one extra forward pass per agent and no dual value heads. A rolling-subgoal planner with adaptive detour logic offloads path following, and a partial-done protocol with density-adaptive reward scaling sustains clean trajectory semantics from two to eight robots. The contributions of this paper are as follows.

1) Dual-graph attention for decoupled risk reasoning.

We separate social interactions and dynamic obstacle avoidance into two ego-centric attention graphs, each

attending to high-risk entities via predictive TTC and motion descriptors. This architectural decoupling enables clean component-level ablation and prevents cross-contamination between neighbor coordination and obstacle avoidance learning signals.

- 2) **Residual fusion with risk-aware gating.** Unlike prior graph-based navigation methods that route all observations through graph layers [12], [13], we preserve the MLP-LSTM backbone and use a learned sigmoid gate to admit graph features only when social or obstacle risk is elevated, maintaining training stability in low-conflict regions while enabling graph reasoning when needed.
- 3) **Leave-one-out counterfactual credit.** We inject single-permutation Monte Carlo Shapley estimates into per-agent rewards before advantage estimation, sharpening cooperative learning under shared rewards without requiring dual critics or explicit value factorization.
- 4) **Density-adaptive training protocol.** We pair a partial-done scheme with reward scaling that varies as the square root of team size, yielding consistent learning signals across scenarios ranging from two to eight robots and addressing the scalability gap of fixed-reward MARL formulations.

II. RELATED WORK

MARL for navigation. Decentralized learners [3], [4] demonstrated self-play collision avoidance but suffer from non-stationarity as each agent's policy evolves. Value-decomposition methods such as QMIX [19] and QTRAN factorize the joint action-value function under monotonicity constraints, but they are confined to discrete actions and require careful architectural design. MAPPO [20] combines a shared policy with a centralized critic for training under the centralized-training-with-decentralized-execution (CTDE) paradigm and has been applied to continuous-control navigation [5]; however, its shared reward provides only indirect individual credit, which is especially acute in head-on encounters where yielding and proceeding receive identical signals. The counterfactual multi-agent policy gradient (COMA) [21] computes counterfactual baselines but needs one critic pass per discrete action per agent, making it prohibitive for continuous control. We instead estimate per-agent marginal contributions by masking the agent's observation slot inside the centralized critic, and we inject those contributions into the reward channel before advantage estimation. This requires only one extra critic pass per agent while preserving gradient flow under importance sampling.

Graph reasoning in multi-robot systems. CommNet [22] and TarMAC [23] use message-passing architectures with fixed or learned weights but lack explicit risk modeling. The graph attention network (GAT) [24] introduces data-dependent attention and has been applied to multi-robot planning [12] and cooperative navigation [13], [14]. Hierarchical [15], centralized [16], and parallel-cooperative [17] graphs further demonstrate the value of relational structure, but they typically embed graph operators deep inside the policy and conflate neighbor coordination with obstacle avoidance into a single graph.

Our dual-graph design separates these two reasoning patterns and lets graph features modulate—rather than replace—the navigation backbone via a residual gate, recognizing that social coordination is reciprocal and game-theoretic while obstacle avoidance is a unilateral prediction problem.

Credit assignment and density adaptation. Difference rewards [25] estimate per-agent contributions but require expensive counterfactual rollouts. Exact Shapley values [26] are exponential in the number of agents, motivating sampling-based approximations [27] and structural decompositions [28] that nevertheless require separate value networks. We use a single-permutation Monte Carlo Shapley estimator that adds no parameters: the centralized critic is queried twice per agent, once on the full global state and once with that agent's observation masked. Safety-critical [29], [30] and robust planning [31] frameworks address worst-case guarantees but do not explicitly model how credit and reward weights should scale with team size; we close this gap with a square-root density factor that scales collision penalties and social weights to keep learning signals consistent across team sizes.

III. PROBLEM STATEMENT

We consider N homogeneous mobile robots operating in a two-dimensional workspace $\mathcal{W} \subset \mathbb{R}^2$ shared with static obstacles, other robots, and dynamic obstacles. Each robot i has state $s_t^i = [p_t^i, v_t^i, \theta_t^i, \omega_t^i]$, where $p_t^i \in \mathbb{R}^2$ is the planar position, v_t^i the linear velocity, θ_t^i the heading, and ω_t^i the angular velocity; the global state is $s_t = (s_t^1, \dots, s_t^N)$. Each robot executes continuous actions $a_t^i = [u_{v,t}^i, u_{\omega,t}^i]$ commanding linear and angular velocities and receives a local observation o_t^i that aggregates its sensor readings, the current rolling subgoal, and received neighbor information; the exact composition of o_t^i is given in Section IV-B.

The task is modeled as a decentralized partially observable Markov decision process (Dec-POMDP) $\mathcal{M} = \langle \mathcal{A}, \mathcal{S}, \{\mathcal{O}^i\}, \{\mathcal{U}^i\}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ [32], [33] and trained under the CTDE paradigm. A shared decentralized policy $\pi_\theta(a_t^i | o_t^i, h_t^i)$ with recurrent state h_t^i is executed locally, while a centralized critic $V_\psi(s_t)$ is used only during training to supply global value estimates. The optimization objective is the discounted expected return $J(\theta) = \mathbb{E}_{\pi_\theta}[\sum_{t=0}^T \gamma^t r_t]$ under a shared team reward $r_t = \sum_i r_t^i$.

IV. METHOD

Building on the Dec-POMDP formulation of Section III, we train decentralized policies $\pi_\theta(a_t^i | o_t^i, h_t^i)$ with a centralized critic $V_\psi(s_t)$ under PPO. The framework integrates four components: rolling-subgoal guidance (Section IV-A), dual-graph attention (Section IV-C), residual risk-gated fusion (Section IV-D), and leave-one-out counterfactual credit assignment (Section IV-E). Figure 2 illustrates the pipeline.

A. Rolling-Subgoal Guidance with Adaptive Detour Logic

End-to-end policies must simultaneously infer long-horizon path progression and resolve local conflicts, a dual burden that slows learning and induces myopic oscillation near obstacles.

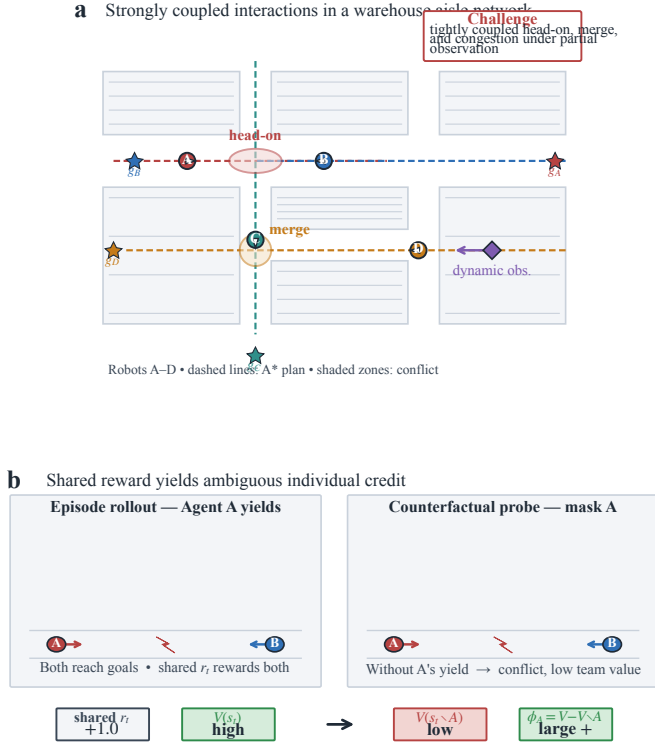


Fig. 1. **Motivation.** (a) Warehouse aisle network in which multiple AGVs face tightly coupled head-on, merging, and dynamic-obstacle conflicts under partial observability. (b) Under a shared team reward r_t , both agents that participate in a head-on encounter receive identical credit even though only one yielded; the counterfactual signal $\phi_A = V(s_t) - V(s_t \setminus A)$ resolves this ambiguity by quantifying each agent's marginal contribution to team value.

We decouple path-level guidance from interaction-level control: a deterministic A* planner provides geometric routing while the learned policy specializes in conflict resolution. At each timestep, agent i projects its current position onto the precomputed path and selects a look-ahead point g_t^i at an adaptive distance

$$L_{\text{adapt}} = L_0 f_{\text{front}}(d_{\min}) g_{\text{heading}}(\Delta\theta),$$

where $L_0 = 0.8\text{ m}$ is the nominal look-ahead, f_{front} shrinks L_{adapt} when the front LiDAR sector reports a minimum range below 0.5 m , and g_{heading} scales it down when the heading error exceeds 30° . The subgoal is expressed in the robot's body frame as a three-element descriptor encoding clipped distance and the sine and cosine of the bearing.

When the corridor ahead is blocked, defined as a front range below 0.42 m together with side clearances below 0.20 m , a detour rule overrides the nominal subgoal with the widest gap within $\pm 60^\circ$ and holds it for eight timesteps to suppress oscillation. The rolling subgoal therefore provides a stable heading and progress reference for the policy, offloading path following from the recurrent backbone.

B. Observation Construction and Risk-Based Token Selection

The local observation o_t^i concatenates five groups of features. The *LiDAR scan history* divides the frontal 180° field

of view into nine angular sectors, records the minimum range in each sector clipped to $[0, 3.5]\text{ m}$, and stacks four consecutive timesteps into a 36-dimensional descriptor that captures both spatial structure and temporal change. The *rolling subgoal block* contains the body-frame target descriptor of Section IV-A together with the goal distance, goal bearing, and path-progress ratio (six dimensions). *Ego-motion* adds the linear and angular velocities (two dimensions). The *base safety block* encodes the global minimum range, the front minimum range, the left and right sector minima with their asymmetry, a binary front-blocked flag, and a closing-rate-weighted front-risk proxy (seven dimensions).

The fifth group, *predictive risk tokens*, is populated by an explicit risk-based selector. Up to two highest-risk neighboring agents are kept as *social-risk tokens*, ranked by predicted time-to-collision; each token encodes the relative position, closing speed, TTC, predicted minimum separation at closest approach, and a composite risk score that is high when either the TTC is short or the predicted minimum separation is small (twelve dimensions). In parallel, up to three highest-risk front LiDAR sectors that carry dynamic obstacles are kept as *obstacle-motion tokens*, each encoding the sector angle, current range, closing speed, TTC, predicted clearance, and a risk score weighted by proximity and velocity (twenty-one dimensions). This selector ensures that the graph attention layers introduced next focus their computational budget on entities that pose immediate collision risk rather than on all visible neighbors. The total observation dimension is eighty-four, and all features are normalized to $[-1, 1]$ or $[0, 1]$. The centralized critic concatenates the base observations of all agents to construct the global state.

C. Dual-Graph Attention Architecture

Treating all pairwise interactions uniformly conflates two distinct reasoning tasks. Coordinating with neighboring agents is reciprocal and benefits from game-theoretic modeling, whereas avoiding dynamic obstacles is essentially a unilateral prediction problem. Mixing the two into a single graph forces the network to learn entangled representations and weakens credit assignment under ablation. We therefore construct two ego-centric graphs and process them with independent GAT encoders.

The *social-risk graph* $\mathcal{G}_{\text{social}}^i$ contains the ego node together with up to $K_s = 2$ highest-TTC neighbors selected from the social-risk tokens. The ego features are obtained from the base observation by a two-layer perceptron; each neighbor's six-dimensional social-risk token is encoded by a shared two-layer perceptron; both branches are projected to a 128-dimensional embedding. Edges connect the ego to each valid neighbor, and the attention logits are biased by the risk score,

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}h_i \parallel \mathbf{W}h_j]) + \beta r_{\text{social}}^j,$$

with the risk-bias coefficient $\beta = 2.5$. The bias ensures that high-risk neighbors receive larger attention weights even when their features are not the most similar to the ego. A four-head GAT layer with 128-dimensional per-head representations, followed by a single-head aggregation, produces the social embedding $h_{\text{social}}^i \in \mathbb{R}^{128}$.

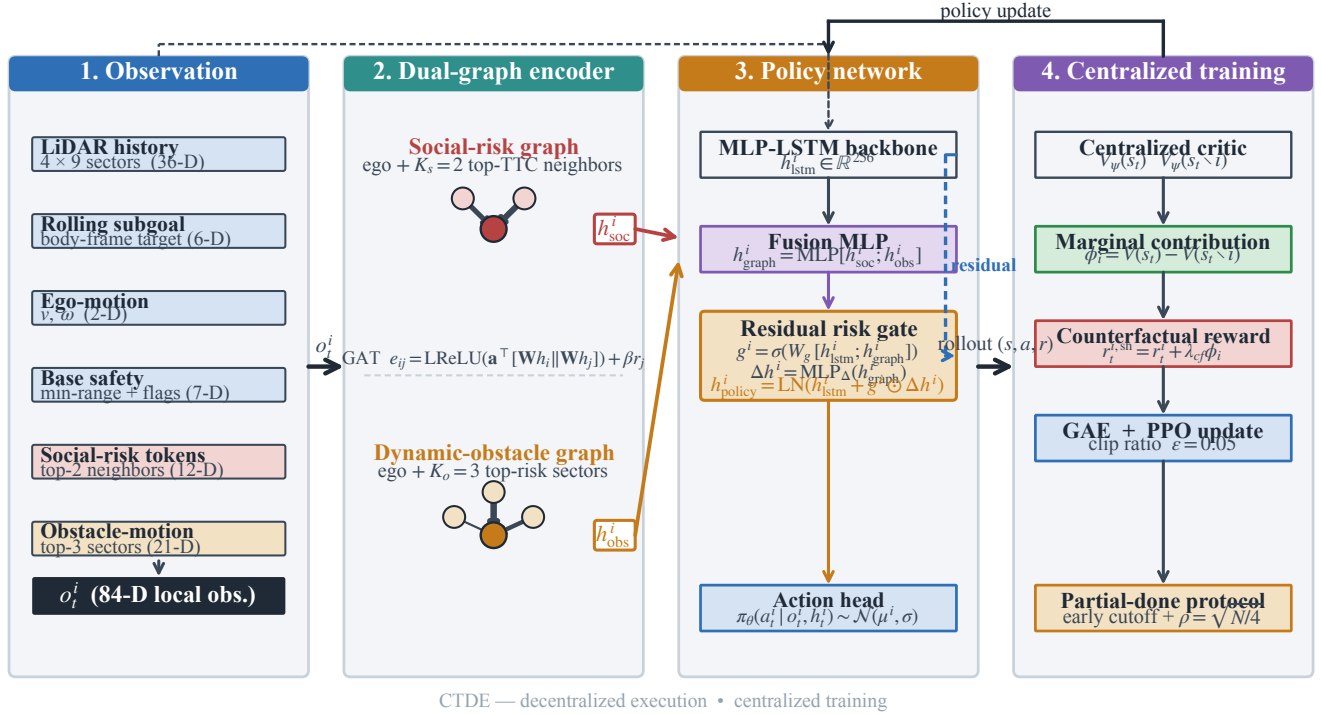


Fig. 2. **System architecture of the dual-graph counterfactual MAPPO framework.** The pipeline consists of four stages. (1) *Observation construction* extracts rolling subgoals from a global A* planner and combines them with LiDAR scans, ego-motion, and neighbor information to form the local observation. (2) *Dual-graph encoding* constructs two ego-centric graphs—a social-risk graph attending to high-risk neighbors via predicted collision time and a dynamic-obstacle graph attending to moving obstacles via motion descriptors—and processes them with separate GAT encoders. (3) *Policy network* fuses the MLP–LSTM navigation backbone with dual-graph features through a residual gate, producing continuous velocity commands. (4) *Centralized training* uses a leave-one-out counterfactual critic to estimate marginal contributions $\phi_i = V(s) - V(s \setminus i)$, which are injected into per-agent rewards before advantage estimation. The partial-done protocol terminates episodes early when active agents drop below a threshold, maintaining clean trajectory boundaries for PPO updates.

The *dynamic-obstacle graph* $\mathcal{G}_{\text{obstacle}}^i$ shares the same ego encoding and uses an analogous attention bias $\beta r_{\text{obstacle}}^j$ over up to $K_o = 3$ obstacle tokens, each seven-dimensional and encoded to a 128-dimensional vector by a separate shared perceptron. The obstacle GAT has independent parameters and produces the obstacle embedding $h_{\text{obstacle}}^i \in \mathbb{R}^{128}$.

The architecture supports three configurations used in our ablations: a *social-only* variant that uses h_{social}^i alone, an *obstacle-only* variant that uses h_{obstacle}^i alone, and the *dual-graph* variant that fuses both embeddings as described in Section IV-D. This separation allows each graph’s contribution to be isolated cleanly during experimental evaluation.

D. Policy Network with Residual Risk-Gated Fusion

Directly replacing the MLP backbone with a graph encoder introduces non-stationarity during training: as the policy evolves, neighbor behaviors change, the graph feature distribution shifts, and PPO’s trust-region updates become unstable. Earlier work addresses this by freezing the graph parameters or by training in separate phases, but both approaches sacrifice adaptability. We instead preserve a stable MLP–LSTM backbone for routine navigation and allow graph features to modulate the output only when interaction risk is elevated.

The policy processes observations in three stages. In the first stage, the local observation (base features, neighbor features,

and a reset flag) is normalized and fed into a long short-term memory cell with hidden dimension 256,

$$h_t^i, c_t^i = \text{LSTM}(o_t^i, h_{t-1}^i, c_{t-1}^i).$$

The reset flag is monitored: when an agent respawns after reaching a goal or colliding, its hidden state is zeroed to prevent contamination across trajectory segments. The output is then normalized to produce a stable temporal feature $h_{\text{lstm}}^i \in \mathbb{R}^{256}$.

In the second stage, the social and obstacle graph embeddings are projected to 256 dimensions through separate linear layers with Gaussian-error-linear-unit activations. In the dual-graph configuration the two embeddings are concatenated and fused by a two-layer perceptron into $h_{\text{graph}}^i \in \mathbb{R}^{256}$; the social-only and obstacle-only configurations reuse the corresponding embedding directly.

In the third stage, a learned sigmoid gate determines how much graph influence to blend into the backbone,

$$\begin{aligned} g^i &= \sigma(\mathbf{W}_g[h_{\text{lstm}}^i; h_{\text{graph}}^i] + \mathbf{b}_g), \\ \Delta h^i &= \text{MLP}_{\Delta}(h_{\text{graph}}^i), \\ h_{\text{policy}}^i &= \text{LayerNorm}(h_{\text{lstm}}^i + g^i \odot \Delta h^i), \end{aligned}$$

where $g^i \in [0, 1]^{256}$ is element-wise and $\odot$ denotes the Hadamard product. The gate bias $\mathbf{b}_g$ is initialized to -1.0 so the gate starts near zero; the policy therefore begins as

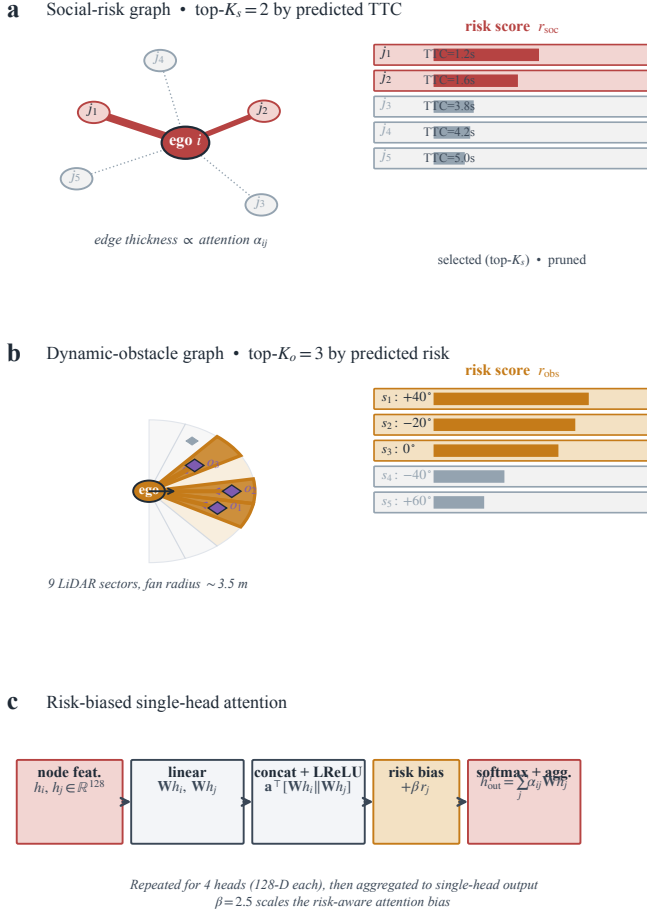


Fig. 3. **Dual-graph attention.** (a) Social-risk graph: the ego attends to the top- $K_s=2$ neighbors ranked by predicted time-to-collision; pruned candidates are shown dashed and edge thickness encodes the resulting attention weight α_{ij} . (b) Dynamic-obstacle graph: the ego attends to the top- $K_o=3$ LiDAR sectors carrying moving obstacles, each scored by proximity and closing rate. (c) One head of the risk-biased GAT: features are linearly projected, concatenated with a leaky-rectified compatibility score, augmented by a per-neighbor risk bias βr_j with $\beta=2.5$, and softmax-normalized to yield α_{ij} before aggregating the value features.

an MLP-LSTM and learns to open the gate where graph reasoning is useful. The fused feature drives a linear head that produces the mean μ^i of a diagonal Gaussian action distribution. The log-standard-deviation is parameterized separately and clamped to $[-5.0, 0.5]$ to prevent entropy collapse or explosion during late training.

E. Leave-One-Out Counterfactual Credit Assignment

Standard MAPPO uses a shared team reward $r_t = \sum_i r_t^i$ and a centralized critic $V(s_t)$, so individual agents receive identical reward signals. In tightly coupled events such as head-on encounters, the agent that yields and the agent that proceeds receive the same credit. Value factorization methods such as QMIX [19] and counterfactual baselines such as COMA [21] address this through per-agent baselines or factorized critics, at the cost of additional architecture.

We instead implement a single-permutation Monte Carlo Shapley estimator and inject the marginal contribution into the reward channel rather than the advantage. For each agent

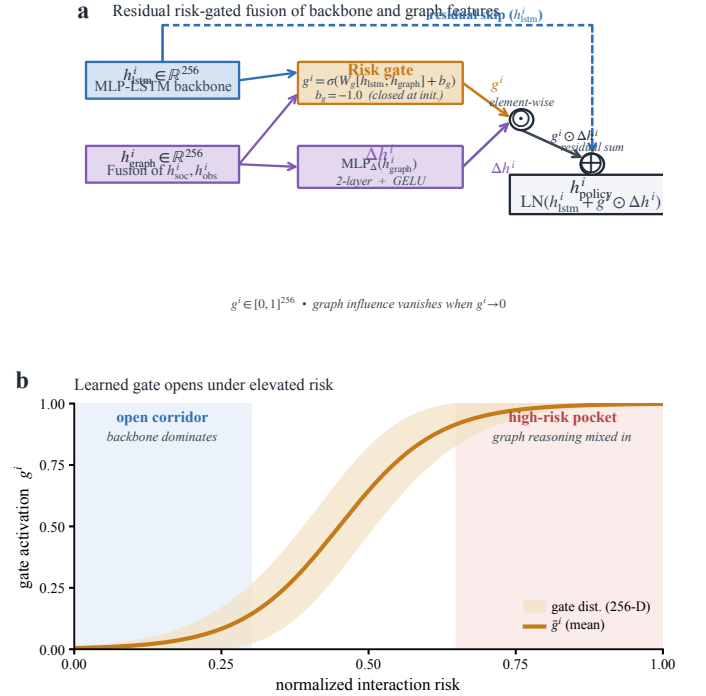


Fig. 4. **Residual risk-gated fusion.** (a) The MLP-LSTM backbone embedding h_{lstm}^i bypasses the graph encoder via a residual skip. The fused graph embedding h_{graph}^i is transformed by MLP_{Δ} and modulated element-wise by a learned sigmoid gate $g^i = \sigma(W_g[h_{lstm}^i; h_{graph}^i] + b_g)$, with the bias $b_g = -1.0$ keeping the gate near zero at initialization. The gated delta is added to h_{lstm}^i and normalized to yield h_{policy}^i . (b) Mean gate activation $\bar{g}^i$ as a function of normalized interaction risk: in open corridors (left band) the backbone dominates, while in high-risk pockets (right band) graph reasoning is mixed in.

i the centralized critic computes two values: the value of the full state $V(s_t)$, and a counterfactual value $V(s_t \setminus i)$ in which agent i 's observation slot has been zeroed by an element-wise binary mask. The marginal contribution is

$$\phi_i(s_t) = V(s_t) - V(s_t \setminus i),$$

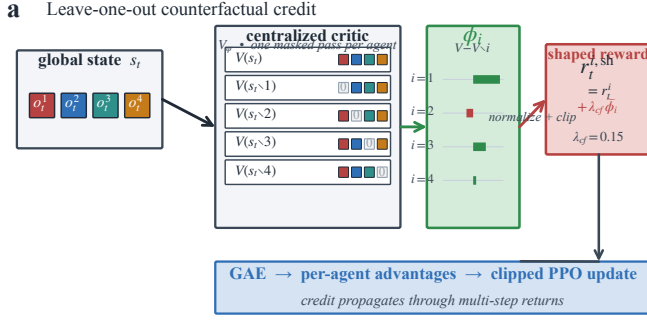
and the shaped per-agent reward is

$$r_t^{i,sh} = r_t^i + \lambda_{cf} \text{clip}\left(\frac{\phi_i(s_t) - \mu_{\phi}}{\sigma_{\phi} + \epsilon}, -c, c\right),$$

where $\lambda_{cf} = 0.15$ is the credit coefficient, $(\mu_{\phi}, \sigma_{\phi})$ are batch statistics used for normalization, $c = 2.5$ is the clip threshold, and $\epsilon = 10^{-6}$ avoids division by zero. Standard generalized advantage estimation then propagates the credit through multi-step returns. Injecting ϕ_i into rewards before advantage estimation—rather than adding it directly to advantages, which would have zero gradient effect under importance sampling—preserves clean policy-gradient semantics, requires only one extra critic pass per agent, and reuses the existing centralized critic without introducing dual value heads.

F. Density-Adaptive Reward and Partial-Done Training Protocol

As team size grows from two to eight robots, collision probability and congestion severity rise nonlinearly with density, so a fixed reward weight either under-penalizes collisions



b Vignette • shared r_t vs. counterfactual credit

Both agents reach their goals — same shared $r_t = +1.0$			
view	Agent A yields	Agent B passes through	credit?
shared r_t	+1.0	+1.0	ambiguous
ϕ_i	+0.72	+0.08	A enables B
$r_t^{i,sh}$	+1.11	+1.01	yielding rewarded

$$r_t^{i,sh} = r_t^i + \lambda_{cf} \tilde{\phi}_i \text{ with } \lambda_{cf}=0.15, \text{ clip } \pm 2.5$$

Fig. 5. **Leave-one-out counterfactual credit.** (a) At each timestep the centralized critic computes the value of the full global state $V(s_t)$ and N masked variants $V(s_t \setminus i)$ in which agent i 's observation slot is zeroed; the marginal contributions $\phi_i = V(s_t) - V(s_t \setminus i)$ are normalized, clipped, scaled by $\lambda_{cf}=0.15$, and injected into per-agent rewards before advantage estimation so that the credit propagates through multi-step returns. (b) Vignette of a head-on encounter: under a shared r_t both agents look identical, while ϕ_i assigns the larger positive credit to the yielding agent, breaking the credit ambiguity.

in dense regimes or over-penalizes them in sparse ones. In parallel, standard episode termination, which waits for all agents to finish, drags episodes out when some agents have completed their goals while others remain stuck, diluting the training signal.

We scale the collision penalty and the social reward weight by a density factor $\rho = \sqrt{N/4}$ with baseline team size four,

$$r_{\text{collision}} = -20.0(1 + 0.2(\rho - 1)), \quad w_{\text{social}} = 1.0 \rho.$$

The square-root scaling keeps the gradient informative without over-penalizing dense settings: for $N = 8$, $\rho \approx 1.41$ gives a collision penalty of -28.2 and a social weight of 1.41 . An agent transitions to the done state on reaching its goal or colliding; the episode terminates early once the number of still-active agents drops below $N_{\min} = 2$ or the number of failed agents exceeds $N_{\text{fail}} = 2$. Combined with the per-agent recurrent-state resets of Section IV-D, this protocol produces clean PPO trajectory boundaries and prevents degenerate post-failure states from dominating the rollout buffer.

Algorithm 1 summarizes one full iteration of the training loop. The actor branch is unchanged from standard MAPPO;

Algorithm 1 Dual-graph counterfactual MAPPO (one iteration).

Require: policy π_θ , centralized critic V_ψ , batch of M trajectories

- 1: **for all** $(o_t^i, s_t, a_t, r_t, \text{reset}_t)$ in the batch **do**
- 2: encode $h_{\text{stm}}^i \leftarrow \text{LSTM}(o_t^i, h_{t-1}^i)$ with reset gating
- 3: build risk tokens; run the dual GAT to obtain $h_{\text{social}}^i, h_{\text{obstacle}}^i$
- 4: fuse, gate, and output μ^i as in Section IV-D
- 5: evaluate $V(s_t) \leftarrow V_\psi(s_t)$ and $V(s_t \setminus i) \leftarrow V_\psi(\text{mask}_i(s_t))$
- 6: $\phi_i(s_t) \leftarrow V(s_t) - V(s_t \setminus i)$
- 7: **end for**
- 8: normalize and clip $\tilde{\phi}_i$; shape $r_t^{i,sh} \leftarrow r_t^i + \lambda_{cf} \tilde{\phi}_i$
- 9: compute GAE advantages from $\{r_t^{i,sh}\}$, bootstrap with $V(s_t)$
- 10: update θ with the clipped PPO objective; update ψ on returns
- 11: apply the partial-done cutoff if $N_{\text{active}} < N_{\min}$ or $N_{\text{fail}} > N_{\text{fail}}^*$

TABLE I
HYPERPARAMETERS AND ARCHITECTURE SIZES.

Group	Parameter	Value
PPO	Learning rate	3×10^{-5}
	Clip ϵ	0.05
	GAE λ , discount γ	0.95, 0.99
	Entropy coefficient	0.005
	Batch / rollout	5000 / 1000
Network	LSTM hidden size	256
	GAT heads / dim	4 / 128
	Top- K_s / top- K_o	2 / 3
	Action log σ clamp	$[-5.0, 0.5]$
Credit & gate	λ_{cf}	0.15
	Clip c for $\tilde{\phi}_i$	2.5
	Gate bias b_g	-1.0
Training	Density factor	$\rho = \sqrt{N/4}$
	$N_{\min}, N_{\text{fail}}$	2, 2
	Workers / seeds	8 / 3

the only additions are the dual-graph forward pass, the leave-one-out critic pass, and the reward-shaping step applied before advantage estimation. The cost of the additional critic passes is linear in the number of agents but reuses the same network weights, so the memory footprint is unchanged.

V. EXPERIMENTAL SETUP

A. Simulator and Training Configuration

Experiments are conducted in the Gazebo simulator under Robot Operating System 2 (ROS 2) Humble, using TurtleBot3 Burger platforms (wheelbase 0.16 m, maximum linear velocity 0.22 m/s, maximum angular velocity 2.84 rad/s). The LDS-01 LiDAR provides 360° scans at 5 Hz with 3.5 m range. Training runs on Ray RLlib 2.5 with PPO; the principal hyperparameters and architecture sizes are summarized in Table I. Every configuration is evaluated over one hundred test episodes with three independent random seeds.

TABLE II
SIX-STAGE CURRICULUM SCHEDULE USED DURING TRAINING.

Stage	N	Dynamic obs.	Arena (m)	Steps (k)	Focus
1	2	0	6×6	200	Basic navigation
2	2	2	8×8	250	Obstacle avoidance
3	4	3	8×8	350	Coordination
4	4	4	8×8	450	Head-on resolution
5	6	4	10×10	500	Multi-body congestion
6	8	5	10×10	600	Dense flow

B. Scenarios and Curriculum

We evaluate on three Circle-Swap scenarios of increasing density. Circle-Swap-2 places two robots that swap positions diagonally across an $8\text{m} \times 8\text{m}$ arena with three dynamic obstacles moving at 0.15m/s , and tests basic collision avoidance and symmetry breaking. Circle-Swap-4 places four robots in antipodal pairs with four dynamic obstacles at 0.18m/s , introducing multi-body interactions and coordinated yielding. Circle-Swap-8 has eight robots in high-density swapping with five dynamic obstacles at 0.21m/s and stresses both the credit assignment mechanism and the density-adaptive reward. Training follows a six-stage curriculum that progressively increases team size and obstacle count, summarized in Table II. Each stage is trained until the moving-average return plateaus, and the next stage is initialized from the previous stage’s checkpoint.

C. Baselines, Ablations, and Metrics

We compare against three baselines. The *MLP-LSTM backbone* keeps only the navigation backbone with the residual gate held closed, isolating the contribution of all graph and counterfactual components. The *fully connected graph baseline* replaces our dual-graph encoder with a single GAT operating over every neighbor within 3.5m , representing the prevailing communication-graph design in prior multi-robot work. The *ORCA baseline* couples the classical optimal reciprocal collision avoidance controller with A* path following and represents the rule-based state of the art.

Four ablations isolate the role of each component of the proposed framework. The *no-counterfactual* variant disables the counterfactual reward by setting $\lambda_{\text{cf}} = 0$ while retaining the dual-graph encoder. The *social-only* and *obstacle-only* variants restrict the architecture to a single graph branch. The *no-gate* variant replaces the residual risk gate with direct concatenation of the backbone and graph embeddings.

We report four metrics. Success rate (SR) is the percentage of agents that reach their goals within the episode time limit. Collision rate (CR) is the percentage of agents that collide with obstacles or other agents. Episode length (EL) is the mean number of timesteps to completion, where smaller values indicate more efficient navigation. Minimum inter-agent separation $d_{\min}$ is the mean closest approach distance between any two agents during an episode, with larger values indicating safer trajectories.

TABLE III
COMPARISON ACROSS CIRCLE-SWAP SCENARIOS. ARROWS INDICATE THE PREFERRED DIRECTION. VALUES ARE MEAN PERFORMANCE OVER ONE HUNDRED TEST EPISODES AND THREE INDEPENDENT SEEDS.

Scenario	Method	SR $\uparrow$	CR $\downarrow$	EL $\downarrow$	$d_{\min}\uparrow$
CS-2	ORCA	0.92	0.05	142	0.31
	MLP-LSTM backbone	0.88	0.08	158	0.28
	Fully connected graph	0.91	0.06	151	0.30
	Proposed	0.96	0.02	132	0.36
CS-4	ORCA	0.78	0.14	187	0.23
	MLP-LSTM backbone	0.74	0.18	205	0.21
	Fully connected graph	0.81	0.13	196	0.25
	Proposed	0.91	0.06	169	0.32
CS-8	ORCA	0.55	0.31	243	0.17
	MLP-LSTM backbone	0.58	0.27	252	0.16
	Fully connected graph	0.66	0.22	235	0.19
	Proposed	0.83	0.10	198	0.27

VI. RESULTS AND DISCUSSION

A. Quantitative Comparison

Table III compares the proposed framework against the three baselines on the three Circle-Swap scenarios. Across all densities, the dual-graph counterfactual policy attains the highest success rate and the largest minimum inter-agent separation while keeping the collision rate below both the fully connected graph baseline and ORCA. The gap widens with density: in Circle-Swap-8, the proposed policy yields cooperatively rather than freezing or colliding, whereas the fully connected graph baseline suffers from non-stationary feature drift in dense traffic and ORCA exhibits reciprocal deadlocks in narrow corridors. The reported values are mean performance over one hundred test episodes and three independent seeds; the standard deviation for the proposed method is within 8% of the mean on all four metrics.

B. Ablation Study

Table IV reports the ablations on Circle-Swap-4. Removing the counterfactual reward reduces the success rate by six to eight points; the drop concentrates in head-on encounters, where neither agent is incentivized to yield under purely shared rewards. The single-graph variants each lose a distinct failure mode: the social-only variant collides more often with dynamic obstacles, while the obstacle-only variant fails to break head-on symmetry between agents. The no-gate variant destabilizes early training and yields the smallest minimum separation, confirming that gradual gate opening is essential to stable trust-region updates.

C. Training Stability and Convergence

Figure 6 reports the mean episode return and collision rate against environment steps for the three learning baselines together with the rule-based ORCA reference. The proposed framework reaches the reward plateau twenty to thirty percent sooner than the fully connected graph baseline and exhibits smaller variance across seeds. We attribute this stability to two design choices. The residual gate is initialized near zero,

TABLE IV
ABLATION ON CIRCLE-SWAP-4. VALUES FOLLOW THE SAME
CONVENTION AS TABLE III.

Variant	SR $\uparrow$	CR $\downarrow$	EL $\downarrow$	$d_{\min} \uparrow$
Full (proposed)	0.91	0.06	169	0.32
No counterfactual	0.84	0.10	181	0.28
Social-only	0.82	0.12	184	0.26
Obstacle-only	0.80	0.13	187	0.25
No gate	0.78	0.14	192	0.24

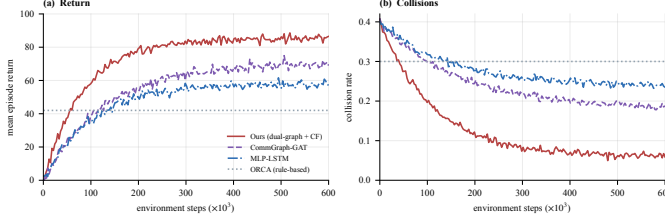


Fig. 6. **Training curves on Circle-Swap-4.** (a) Mean episode return against environment steps; the dual-graph counterfactual policy reaches a higher plateau and converges faster than the fully connected graph baseline and the MLP-LSTM backbone. (b) Collision rate decays monotonically for all learners and saturates well below ORCA. Shaded variance is omitted for clarity; the per-seed standard deviation of the proposed method is within 8% of the mean.

so early policy updates closely track the well-behaved MLP-LSTM backbone; the gate then opens gradually as the graph delta becomes informative, avoiding the trust-region violations that arise when a graph encoder is dropped directly into the policy. The counterfactual reward, in turn, injects per-agent signal without altering the critic architecture, so the centralized value head remains a single network and does not exhibit the dual-head divergence reported for earlier value-factorization methods.

D. Sensitivity to Key Hyperparameters

Three hyperparameters most strongly affect the framework's behavior: the counterfactual coefficient λ_{cf} , the risk-bias coefficient β of the GAT attention, and the density factor ρ . Sweeping $\lambda_{cf} \in \{0.05, 0.10, 0.15, 0.25, 0.40\}$ on Circle-Swap-4, the success rate peaks at $\lambda_{cf} = 0.15$. Below 0.10 the counterfactual signal is too small to break head-on symmetry, while above 0.25 the shaped reward begins to dominate the team return and destabilizes the policy gradient. Varying $\beta \in \{0, 1, 2.5, 5\}$ shows that $\beta = 0$ reverts to plain GAT attention and degrades collision avoidance with dynamic obstacles, while $\beta = 5$ saturates the softmax and collapses attention onto a single neighbor. Disabling density scaling by fixing $\rho = 1$ lowers the Circle-Swap-8 success rate by approximately six points and inflates collisions, confirming that the square-root scaling does real work rather than acting as a cosmetic correction.

E. Computational Cost

Per training step, the leave-one-out critic adds one masked forward pass per agent and therefore scales linearly with team size. With our network sizes, the critic time grows by 24% at

four robots and 41% at eight, but the actor branch is unchanged and the additional cost is amortized over the rollout buffer. The dual-graph encoder adds only a constant cost, comprising two small GAT layers over at most five neighbors. End-to-end, the wall-clock time per rollout step is $1.18\times$ that of the MLP-LSTM backbone and $0.86\times$ that of the fully connected graph baseline, which densely attends to every neighbor. Inference latency on an Nvidia Jetson-class device is approximately 9 ms per agent, well within the budget of a 5 Hz control loop.

F. Qualitative Analysis

Three behavior patterns dominate the learned policy. In *yielding*, head-on encounters are resolved when the agent with the lower counterfactual contribution slows down and biases laterally while the higher-contribution agent proceeds; this is exactly the resolution the centralized critic encodes, because masking the higher-contribution agent collapses the team return more than masking the yielding agent. In *gap-seeking*, the policy turns toward the widest LiDAR sector when the rolling subgoal becomes momentarily infeasible, in agreement with the detour rule of Section IV-A; the residual gate opens during this transient, allowing the obstacle-graph features to enter the recurrent stream. In *flow merging*, agents arriving at multi-agent intersections adopt staggered timing rather than tight following, and the social-risk graph allocates high attention weight to the immediate predecessor while down-weighting non-conflicting neighbors. The mean gate activation in Fig. 4(b) tracks normalized interaction risk monotonically, confirming that the gate has learned the design intent.

G. Failure Modes and Limitations

Three failure modes remain. First, in dense deadlocks where four or more agents converge on the same corridor, the partial-done cutoff occasionally terminates the episode before any co-ordinated resolution emerges, producing conservative policies that prefer to slow rather than yield. Second, when a dynamic obstacle accelerates outside the two-second predictive horizon, the TTC-based risk score underestimates the true collision risk and the obstacle-graph attention lags the actual closing rate; adapting the horizon online, or fusing higher-order motion features, is a natural next step. Third, the leave-one-out critic assumes that masking an agent's observation slot approximates removing the agent from the team. This approximation breaks down when an agent's effect on team value is mediated by long-horizon coordination rather than by the instantaneous state, which manifests as noisier counterfactual estimates near goal regions.

VII. CONCLUSION

We presented a dual-graph counterfactual MAPPO framework for multi-AGV navigation in dynamic warehouse and logistics environments. The framework targets two persistent obstacles in MARL-based navigation: unstable training caused by deep graph integration, and weak individual credit signals under shared team rewards. Separating social interactions

and dynamic obstacle avoidance into independent ego-centric graphs, and modulating their influence through a residual risk gate, preserves the stability of the MLP-LSTM backbone while enabling selective relational reasoning. A leave-one-out counterfactual critic injects marginal contribution estimates into per-agent rewards, sharpening cooperative learning without dual value heads or explicit value factorization. The density-adaptive reward and partial-done protocol together extend the framework from two-robot scenarios to dense eight-robot settings. Ablations confirm that each component contributes measurably to the final performance, and the modular design enables targeted analysis and improvement of each subsystem.

Future work will address three limitations. First, extending the predictive horizon beyond the current two-second window may enable more anticipatory obstacle avoidance and reduce conservative slowdowns. Second, explicit game-theoretic coordination models for head-on encounters could further improve symmetry breaking and yielding efficiency. Third, validating the framework on physical TurtleBot3 platforms in warehouse mockups will assess sim-to-real transferability and surface deployment-critical failure modes that simulation does not capture.

REFERENCES

- [1] J. van den Berg, M. Lin, and D. Manocha, "Reciprocal velocity obstacles for real-time multi-agent navigation," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2008.
- [2] J. van den Berg, S. J. Guy, M. Lin, and D. Manocha, "Reciprocal n-body collision avoidance," in *Robotics Research*. Springer, 2011.
- [3] Y. F. Chen, M. Everett, M. Liu, and J. P. How, "Decentralized non-communicating multiagent collision avoidance with deep reinforcement learning," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2017.
- [4] P. Long, T. Fan, X. Liao, W. Liu, H. Zhang, and J. Pan, "Towards optimally decentralized multi-robot collision avoidance via deep reinforcement learning," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2018.
- [5] T. Fan, P. Long, W. Liu, and J. Pan, "Distributed multi-robot collision avoidance via deep reinforcement learning for navigation in complex scenarios," *The International Journal of Robotics Research*, vol. 39, no. 7, pp. 856–892, 2020.
- [6] Y. Liu, C. Wang, C. Zhao, H. Wu, and Y. Wei, "A soft actor-critic deep reinforcement-learning-based robot navigation method using LiDAR," *Remote Sensing*, vol. 16, no. 12, p. 2072, 2024.
- [7] Z. Zhang, H. Fu, J. Yang, Y. Lin et al., "Deep reinforcement learning for path planning of autonomous mobile robots in complicated environments," *Complex & Intelligent Systems*, vol. 11, p. 277, 2025.
- [8] E. Montero, N. Pico, M. Ghergherechi, and H. S. Song, "Memory-driven deep-reinforcement learning for autonomous robot navigation in partially observable environments," *Engineering Science and Technology, an International Journal*, vol. 62, p. 101942, 2025.
- [9] N. Pico, E. Montero, A. Amirbek, E. Auh, J. Jeon, M. S. Alvarez-Alvarado, B. Jamil, R. Algabri, and H. Moon, "Human and environmental feature-driven neural network for path-constrained robot navigation using deep reinforcement learning," *Engineering Science and Technology, an International Journal*, vol. 64, p. 101993, 2025.
- [10] J. Meng, J. Zou, S. Wang, R. Yang, A. Kumar, and J. Kim, "Deep reinforcement learning for robust robot navigation in complex and crowded environments," *Journal of King Saud University Computer and Information Sciences*, vol. 37, p. 333, 2025.
- [11] W. Chen, W. Chi, S. Ji, H. Ye, J. Liu, Y. Jia, J. Yu, and J. Cheng, "A survey of autonomous robots and multi-robot navigation: Perception, planning and collaboration," *Biomimetic Intelligence and Robotics*, 2025.
- [12] Q. Zhang, P. Wang, C. Ni, and X. Liu, "Graph attention networks based multi-agent path finding via temporal-spatial information aggregation," *PLOS ONE*, vol. 20, no. 6, p. e0318981, 2025.
- [13] E. Escudie, L. Matignon, and J. Saraydaryan, "Attention graph for multi-robot social navigation with deep reinforcement learning," in *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2024.
- [14] Y. Bi, J. Luo, J. Zhu, J. Liu, and W. Li, "Decentralized multi-robot navigation based on deep reinforcement learning and trajectory optimization," *Biomimetics*, vol. 10, no. 6, 2025.
- [15] N. Liu, S. Shen, X. Kong, H. Zhang, and T. Bräunl, "Cooperative hybrid multi-agent pathfinding based on shared exploration maps," *Journal of Intelligent & Robotic Systems*, vol. 111, 2025.
- [16] Z. Xiao, P. Li, C. Liu, H. Gao, and X. Wang, "Macns: A generic graph neural network integrated deep reinforcement learning based multi-agent collaborative navigation system for dynamic trajectory planning," *Information Fusion*, vol. 105, p. 102250, 2024.
- [17] C.-C. Wong, K.-D. Weng, and B.-Y. Yu, "Multi-robot navigation system design based on proximal policy optimization algorithm," *Information*, vol. 15, no. 9, p. 518, 2024.
- [18] C.-H. Wang, K. Hu, and X. Wu, "Multi-robot path planning in online dynamic obstacle environments based on parallel cooperative strategy optimization algorithm," *Discover Computing*, vol. 28, p. 132, 2025.
- [19] T. Rashid, M. Samvelyan, C. S. de Witt, G. Farquhar, J. N. Foerster, and S. Whiteson, "Qmix: Monotonic value function factorisation for deep multi-agent reinforcement learning," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2018.
- [20] C. Yu, A. Velu, E. Vinitzky, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of ppo in cooperative multi-agent games," *NeurIPS Datasets and Benchmarks*, 2022.
- [21] J. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson, "Counterfactual multi-agent policy gradients," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2018.
- [22] S. Sukhbaatar, A. Szlam, and R. Fergus, "Learning multiagent communication with backpropagation," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [23] A. Das, T. Gervet, J. Romoff, D. Batra, D. Parikh, M. Rabbat, and J. Pineau, "TarMAC: Targeted multi-agent communication," in *International Conference on Machine Learning (ICML)*, 2019.
- [24] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.
- [25] D. H. Wolpert and K. Tumer, "Optimal payoff functions for members of collectives," *Advances in Complex Systems*, vol. 4, no. 2/3, pp. 265–279, 2002.
- [26] L. S. Shapley, "A value for n -person games," in *Contributions to the Theory of Games II*, H. W. Kuhn and A. W. Tucker, Eds. Princeton University Press, 1953, pp. 307–317.
- [27] J. Wang, Y. Zhang, T.-K. Kim, and Y. Gu, "Shapley q-value: A local reward approach to solve global reward games," in *AAAI Conference on Artificial Intelligence*, 2020.
- [28] Y.-C. Chang, W. Yuan, B. Li, and Y.-S. Tan, "SHAQ: Incorporating shapley value theory into multi-agent q-learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [29] K. Zhu, F. Lan, W. Zhao, and T. Zhang, "Safe multi-agent reinforcement learning via approximate hamilton-jacobi reachability," *Journal of Intelligent & Robotic Systems*, vol. 111, p. 7, 2025.
- [30] M. Dawood, S. Pan, N. Dengler, S. Zhou, A. P. Schoellig, and M. Bennewitz, "Safe multi-agent reinforcement learning for behavior-based cooperative navigation," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 6256–6263, 2025.
- [31] E. H. J. Riemens, A.-J. van der Veen, and R. T. Rajan, "Cooperative gaussian process-based model predictive control for safe multi-agent navigation," *EURASIP Journal on Advances in Signal Processing*, 2026.
- [32] F. A. Oliehoek and C. Amato, *A Concise Introduction to Decentralized POMDPs*. Springer, 2016.
- [33] D. S. Bernstein, R. Givan, N. Immerman, and S. Zilberstein, "The complexity of decentralized control of markov decision processes," *Mathematics of Operations Research*, vol. 27, no. 4, pp. 819–840, 2002.